

# Deep Learning for Skin Cancer Image Analysis: A Systematic Literature Review

## Methodology

### Literature Search

*PRISMA 2020 Protocol  
Scopus, IEEE, ACM, Springer, ArXiv*

### Screening

*Quantitative Impact Filter  
DL Eligibility Assessment*

### Included Studies

*n = 254 Primary Studies*

## DL Diagnostic Pipeline

### Data Foundation

*ISIC Archive, HAM10000,  
PH2, PAD-UFES-20*

### Preprocessing & Augmentation

*Artifact Removal, Spatial Transforms &  
GAN Synthesis*

### Architectural Shift

*Foundational CNNs ->  
Vision Transformers (ViTs) & Ensembles*

### Computational Tasks

**Segmentation:** *Lesion & Attributes*  
**Classification:** *Binary & Multi-class*

## Clinical Translation

### Persistent Challenges

- Severe Data Imbalance
- Poor OOD Generalization
- Opaque AI (Black-Box Models)
- High Computational Overhead

### Future Directions

- Explainable AI (XAI)
- Multimodal Patient Data Integration
- Federated Learning (FL)
- Real-World Clinical Validation